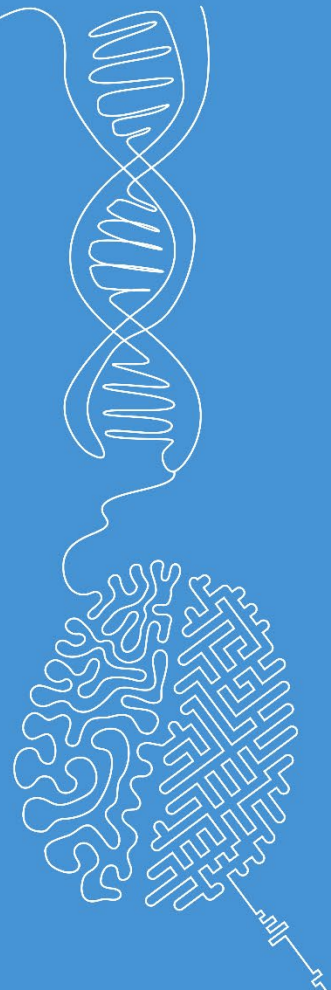


PET / SPECT

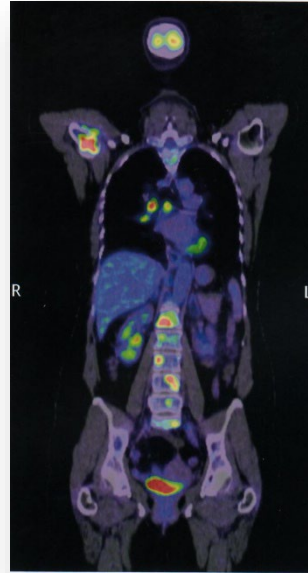
Imaging for the Life Sciences

Robert Vorburger



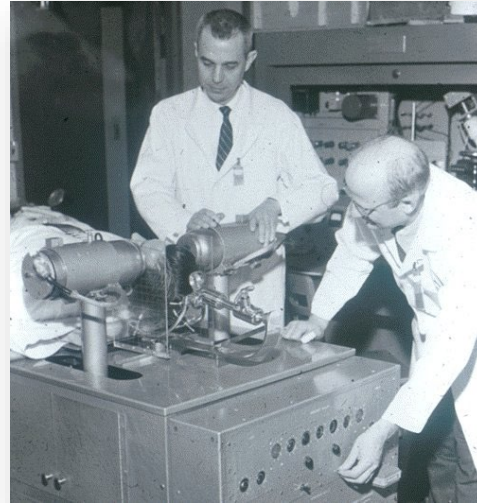
Content

- History
- Principles
- Applications
- Data Aspects

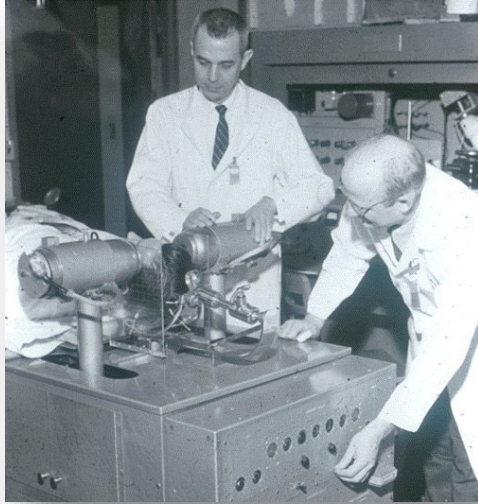


Content

- **History**
- Principles
- Applications
- Data Aspects



Developments in the 1950s and 1960s



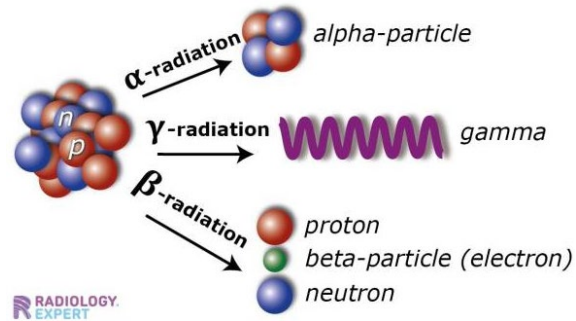
The first PET scanner in the Physics Research Laboratory at Massachusetts General Hospital (1953)



The first digital whole-body SPECT scanner (homemade). Dr Ian Keyes is operating the machine at the Royal Infirmary, Aberdeen (1968)

Content

- History
- Principles
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Nuclear Radiation

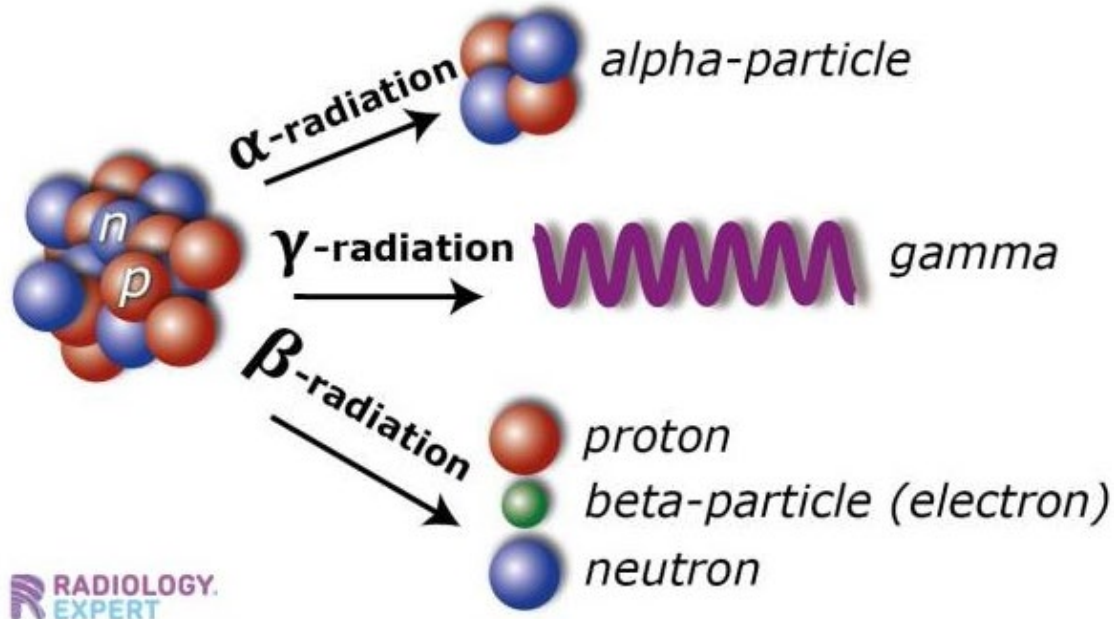
Imaging with nuclear radiation...



source: modernsurvivalblog.com

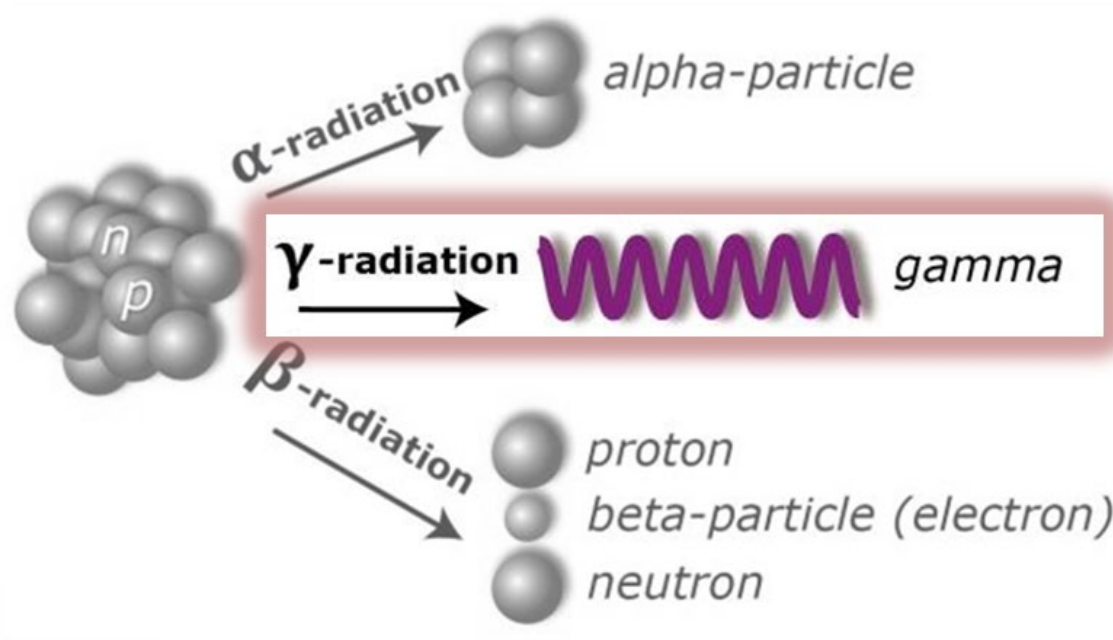
Nuclear Radiation

Imaging with nuclear radiation... **at very small doses**



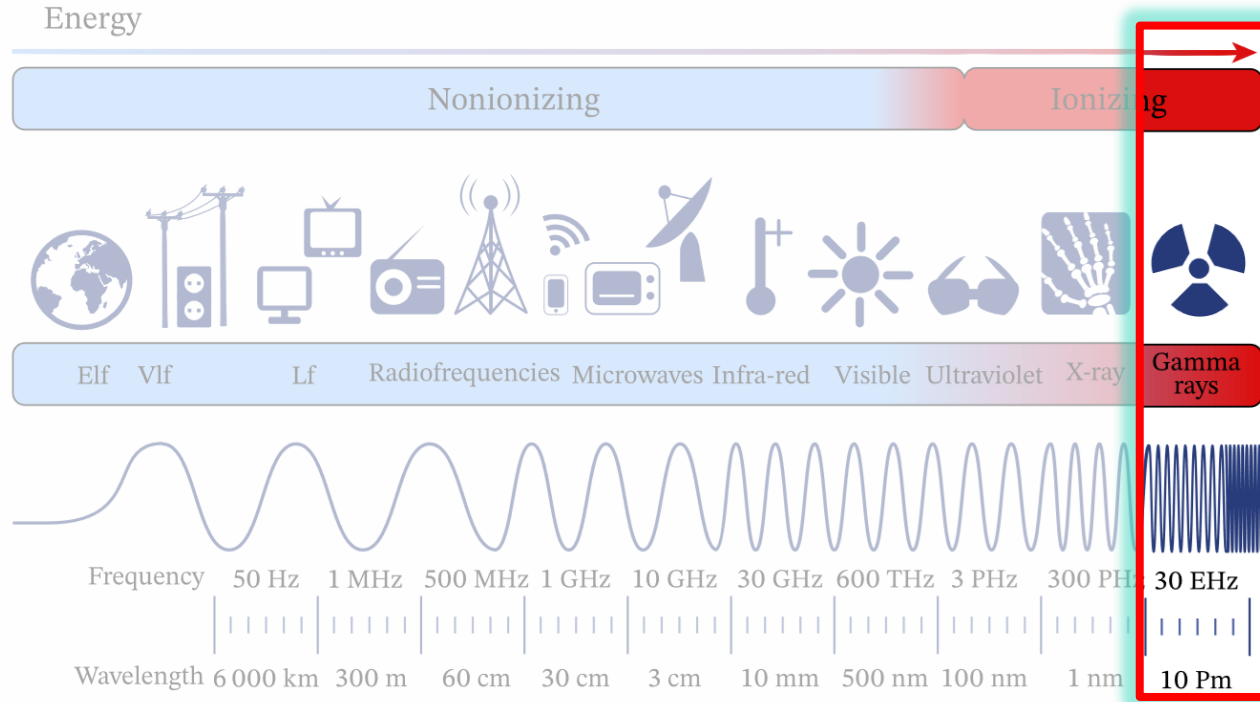
source: www.radiology.expert

Gamma Rays



source: www.radiology.expert

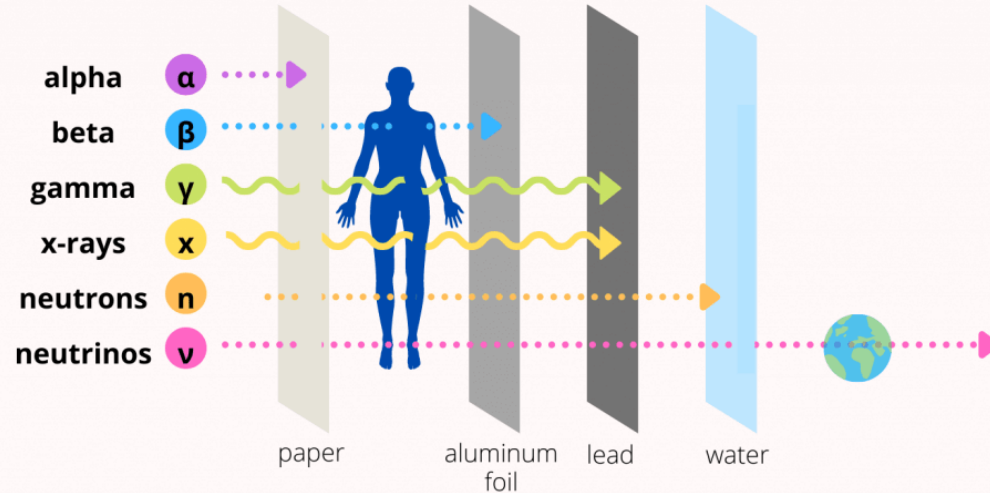
Gamma Rays



source: nagwa.com

Gamma Rays

Which Type of Radiation Is the Most Penetrating?



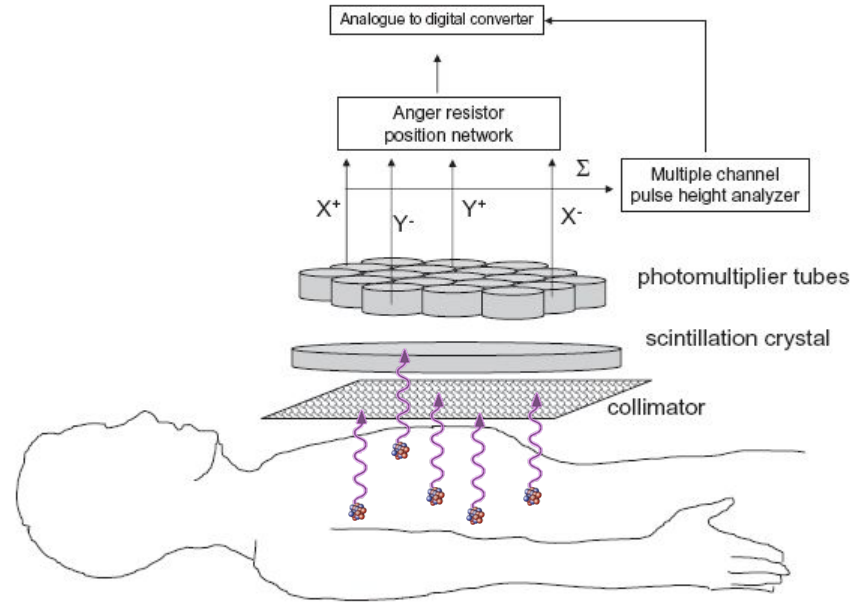
Gamma rays have the most penetrating power of common types of radiation. But, neutrinos have the most penetrating power of all.

sciencenotes.org

Planar scintigraphy

Production of an image (planar projection) with a «gamma camera»:

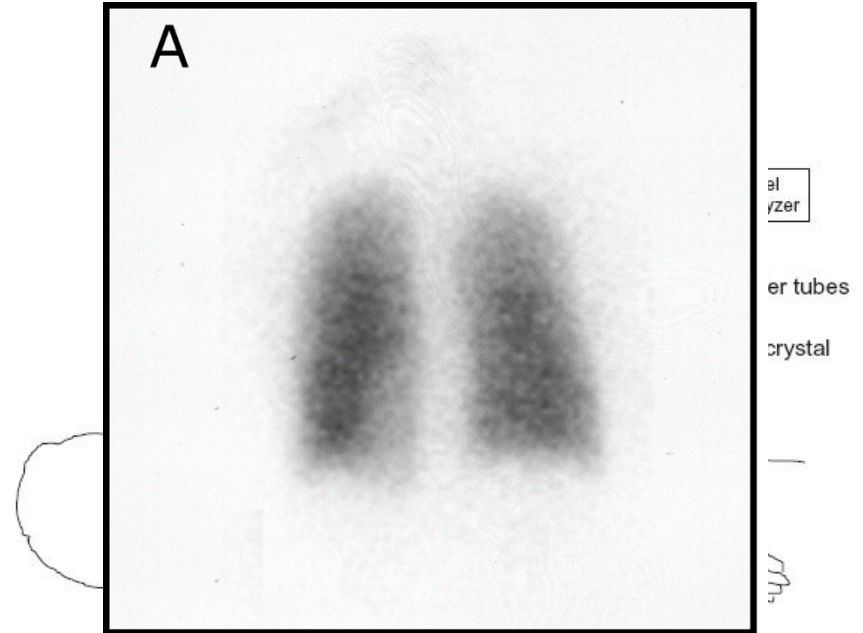
1. Radiotracer emits photons in all directions
2. Photons are scattered within body
3. Collimator: accepts only photons from certain direction («sharpens»)
4. Scintillation crystal: converts gamma-photon into visible photon
5. Photomultiplier tubes: amplifies number of photons
6. Rest: signal enhancement and analogue-to-digital conversion
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Planar scintigraphy

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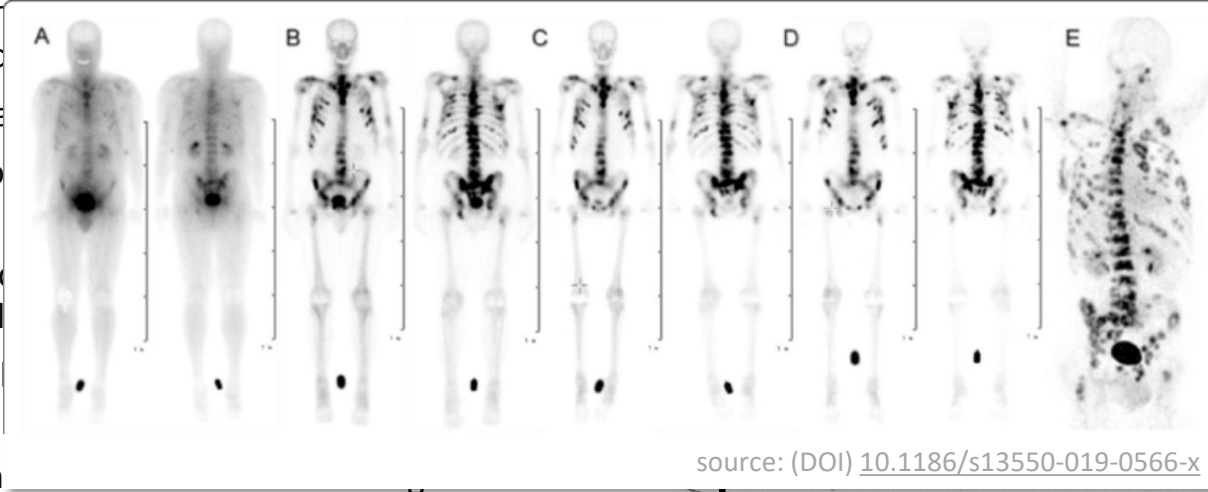
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Planar scintigraphy

Production of an image (planar projection) with a «gamma camera».

1. Radiotracer
2. Photons are emitted
3. Collimator selects photons in a specific direction
4. Scintillation of the crystal into visible light
5. Photomultiplier tubes convert light into electrons
6. Rest: signal processing and digital conversion



A

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er tubes

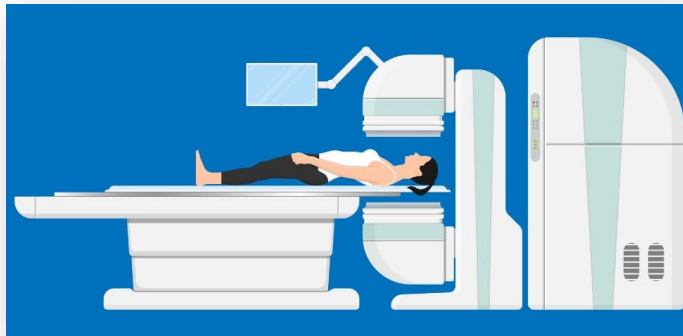
crystal

74

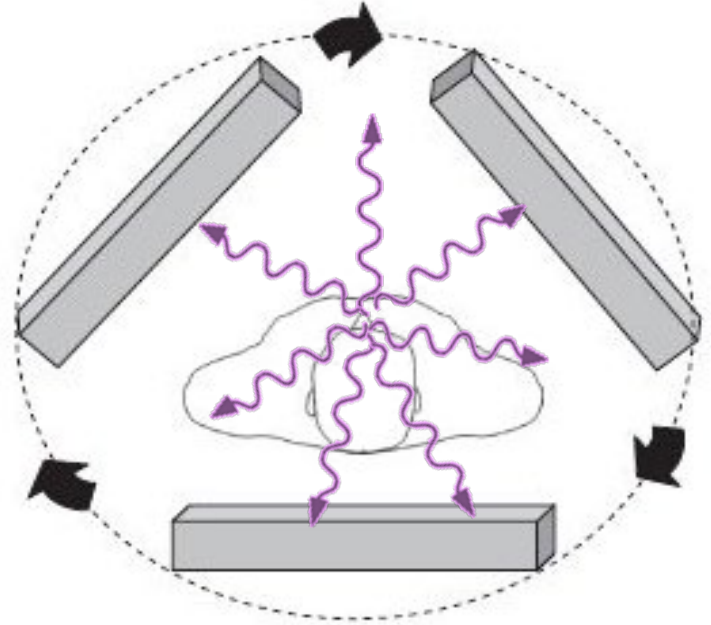
SPECT

Single photon emission tomography (SPECT):

- Tomographic version of planar scintigraphy
- At least two gamma cameras rotate around patient → tomographic reconstruction possible
- Almost always combined with a CT for anatomical information (displayed in the background)



source: healthlibrary.askapollo.com

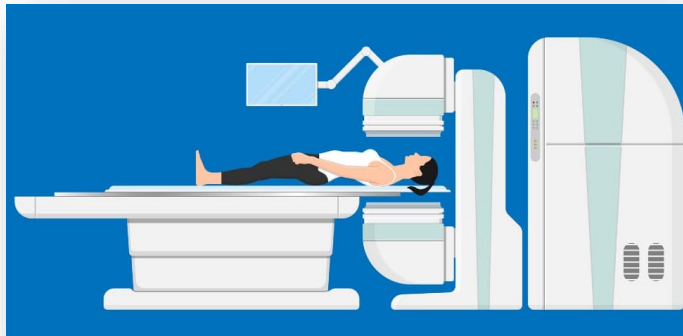


source: Introduction to Medical Imaging, Cambridge University Press, Nadine Barrie Smith, Andrew Webb

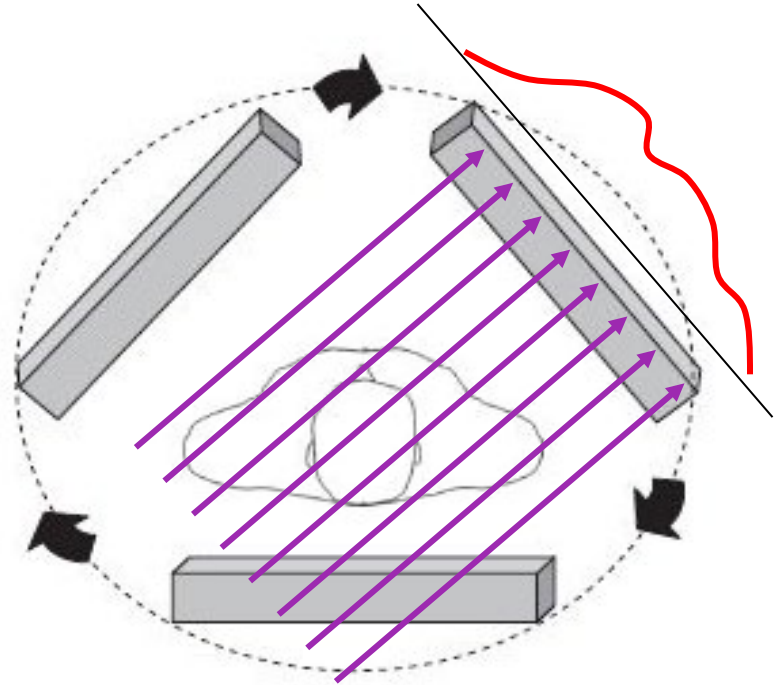
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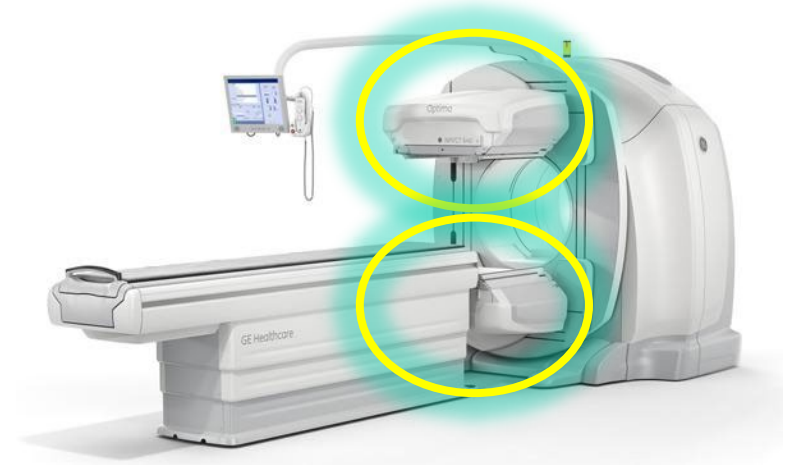


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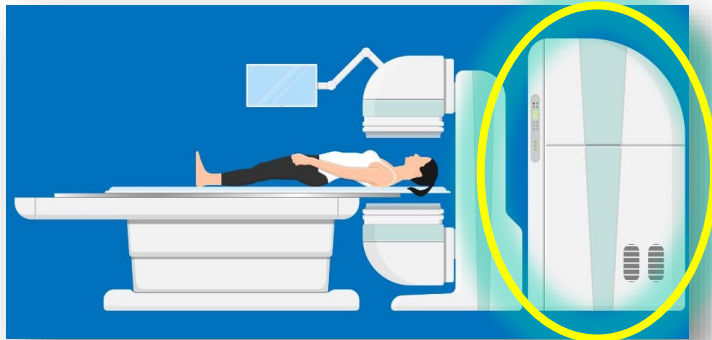
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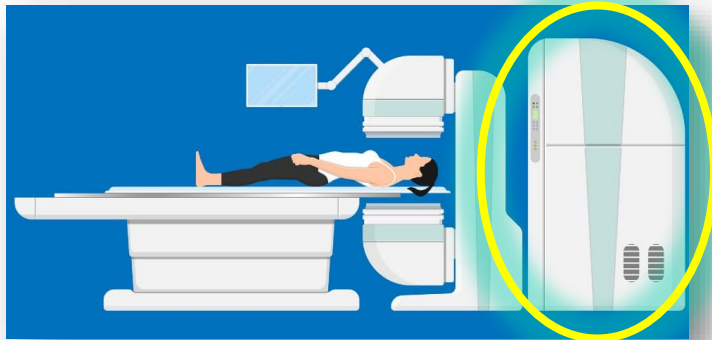
source: healthlibrary.askapollo.com



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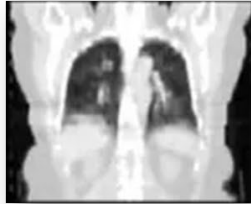
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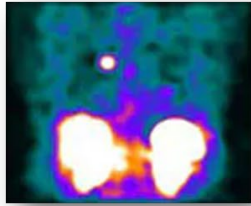
source: healthlibrary.askapollo.com



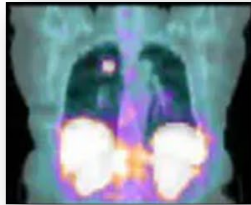
SPECT



CT

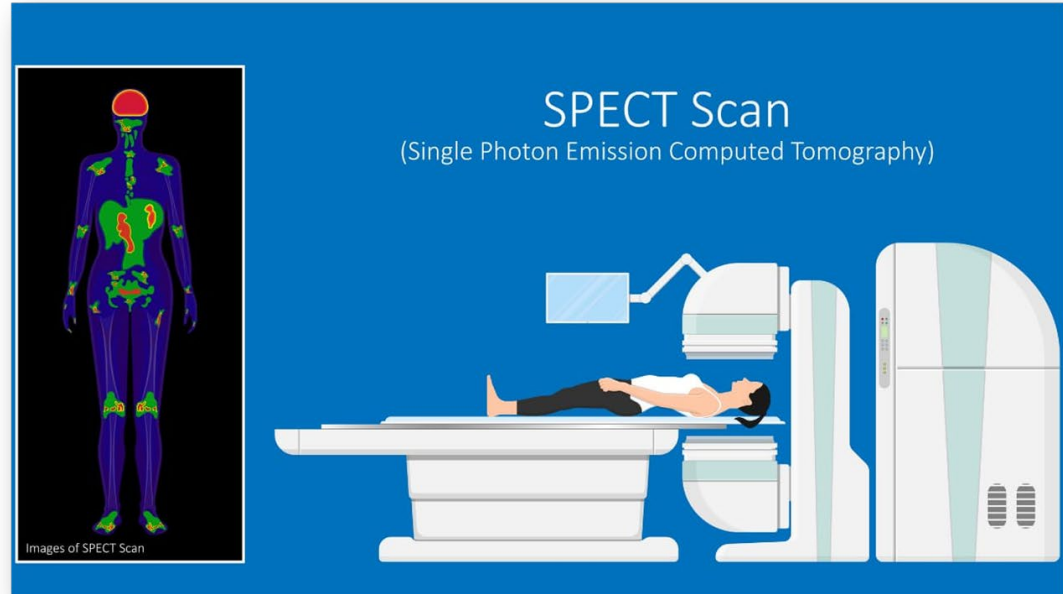


SPECT



CT-SPECT

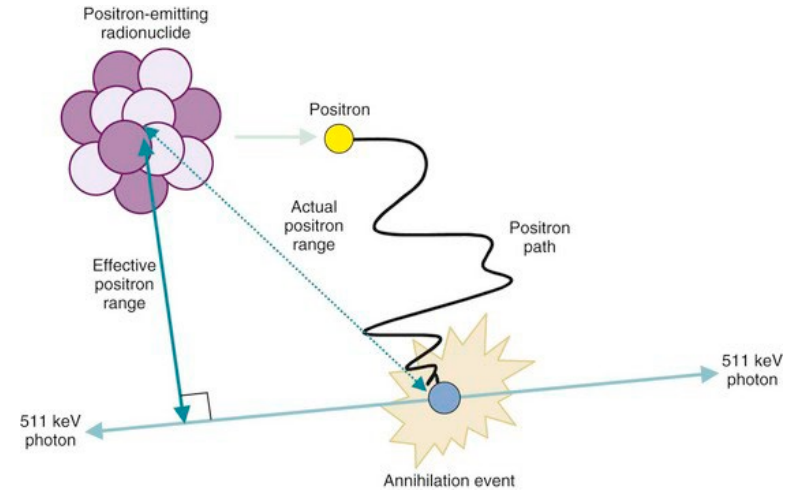
source: mayoclinic.org



PET

Positron emission tomography (PET):

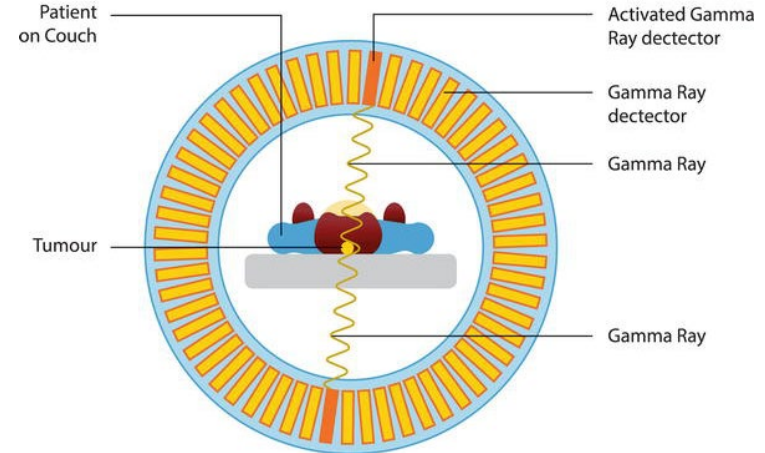
- Radiotracers emits positron (e^+)
(no direct gamma rays)
- Positron (e^+) annihilates by hitting an electron (e^-) in tissue and emits two Gamma-rays, going in opposite direction
- Production of radiotracers requires expensive equipment (cyclotron)



PET

Positron emission tomography (PET):

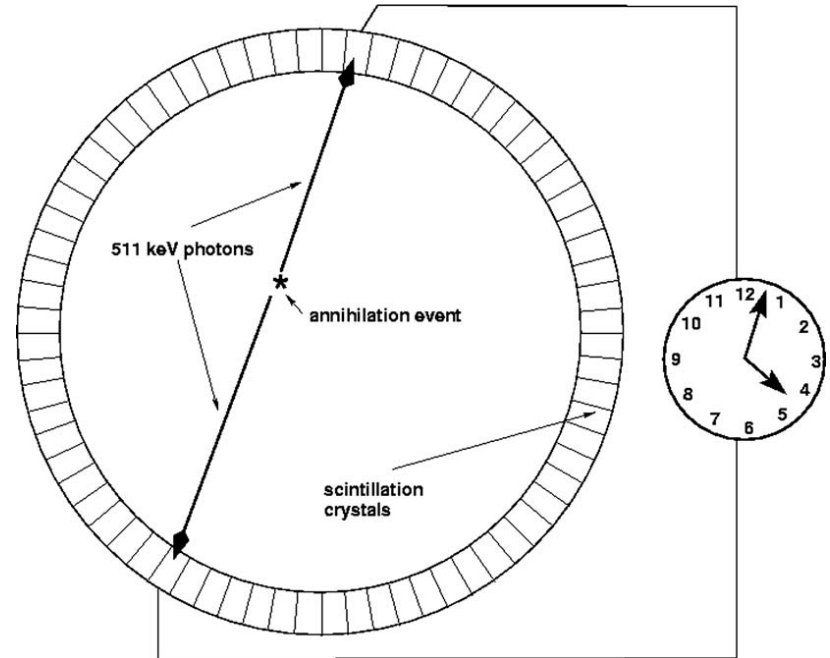
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- Complete ring of detectors $\rightarrow$ tomographic reconstruction possible
- No collimation needed because of coincidence detection for spatial resolution
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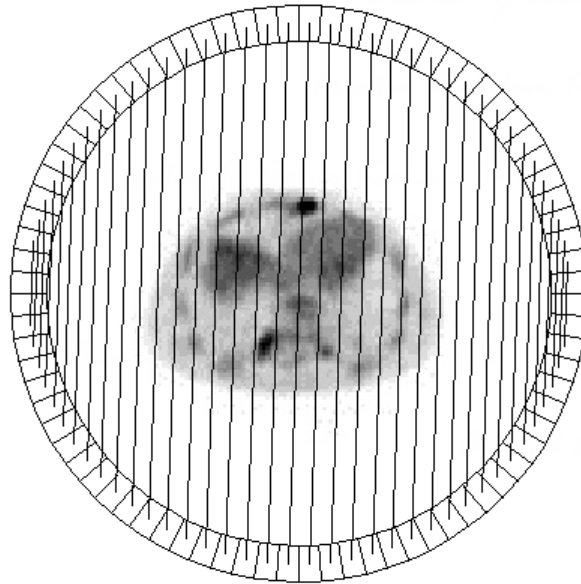
PET

 [animation online](#)

Positron emission tomography

- Radiotracers emits γ rays (no direct gamma rays) (no direct gamma rays)
- Positron (e^+) annihilation and emits two Gamma rays
- Production of radiotracers (cyclotron)
- Complete ring of detectors possible
- No collimation needed for spatial resolution
- Almost always combined with CT (displayed in the background)

Lines of response between PET detectors



Angle: 0°

Corresponding location in sinogram



Rho (offset)

IAEA Human Health Campus — humanhealth.iaea.org

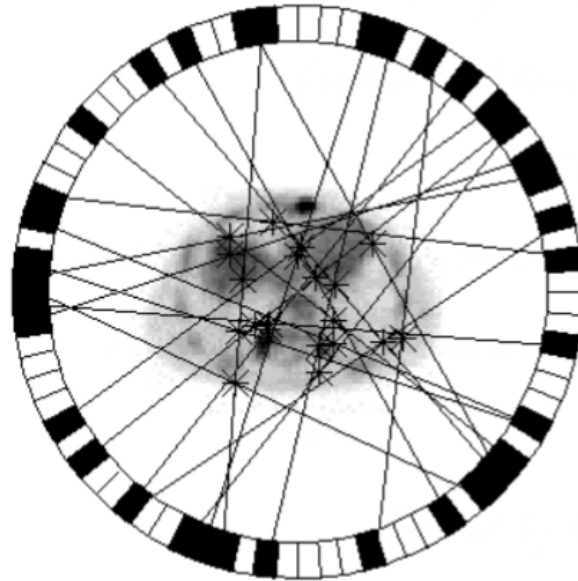
PET

 [animation online](#)

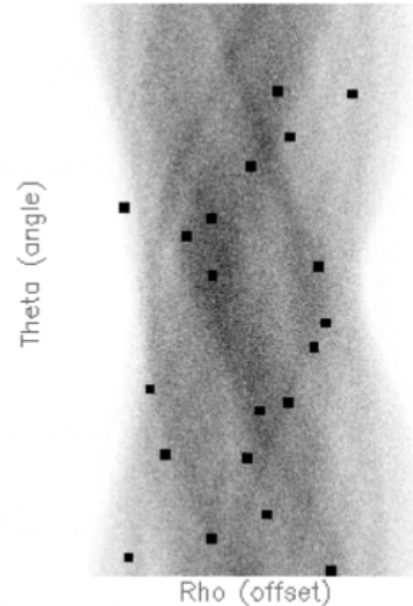
Positron emission tomography

- Radiotracers emits
- Positron (e^+) annihilates with electron in tissue and emits two gamma rays in opposite directions
- Production of radioactive tracers requires specialized equipment (cyclotron)
- Complete ring of detectors allows for full 3D reconstruction possible
- No collimation needed for detection for spatial localization
- Almost always combined with CT or MRI for anatomical information (displayed in color)

Emission volume + PET detectors



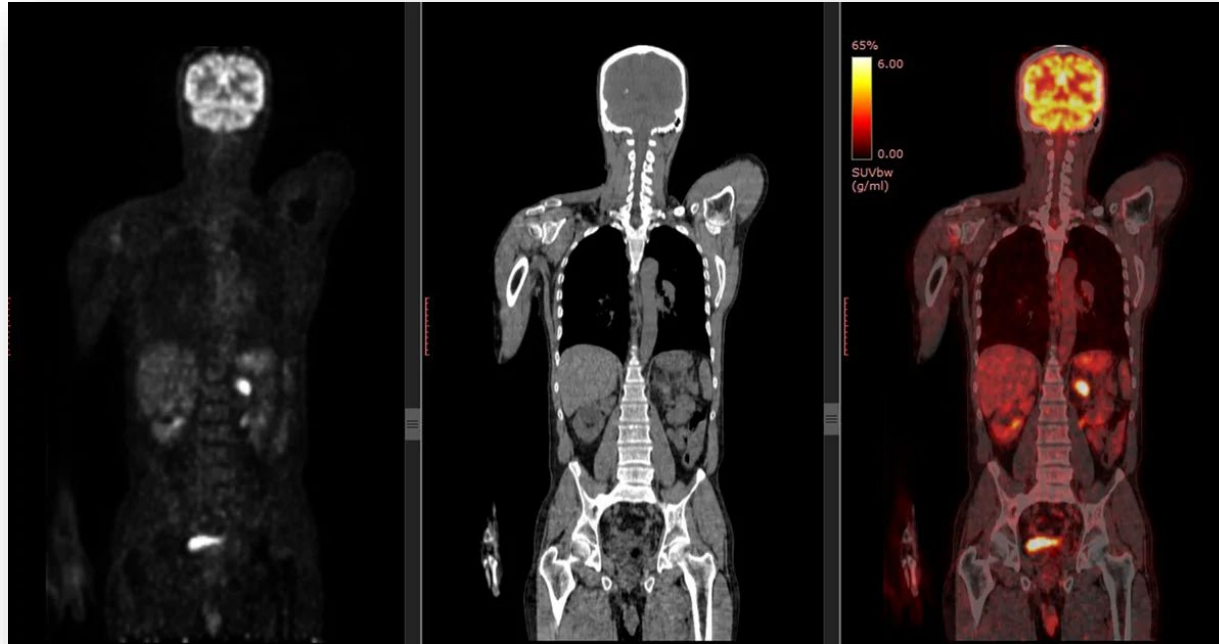
Sinogram (histogram)



Events processed: 16670

IAEA Human Health Campus — humanhealth.iaea.org

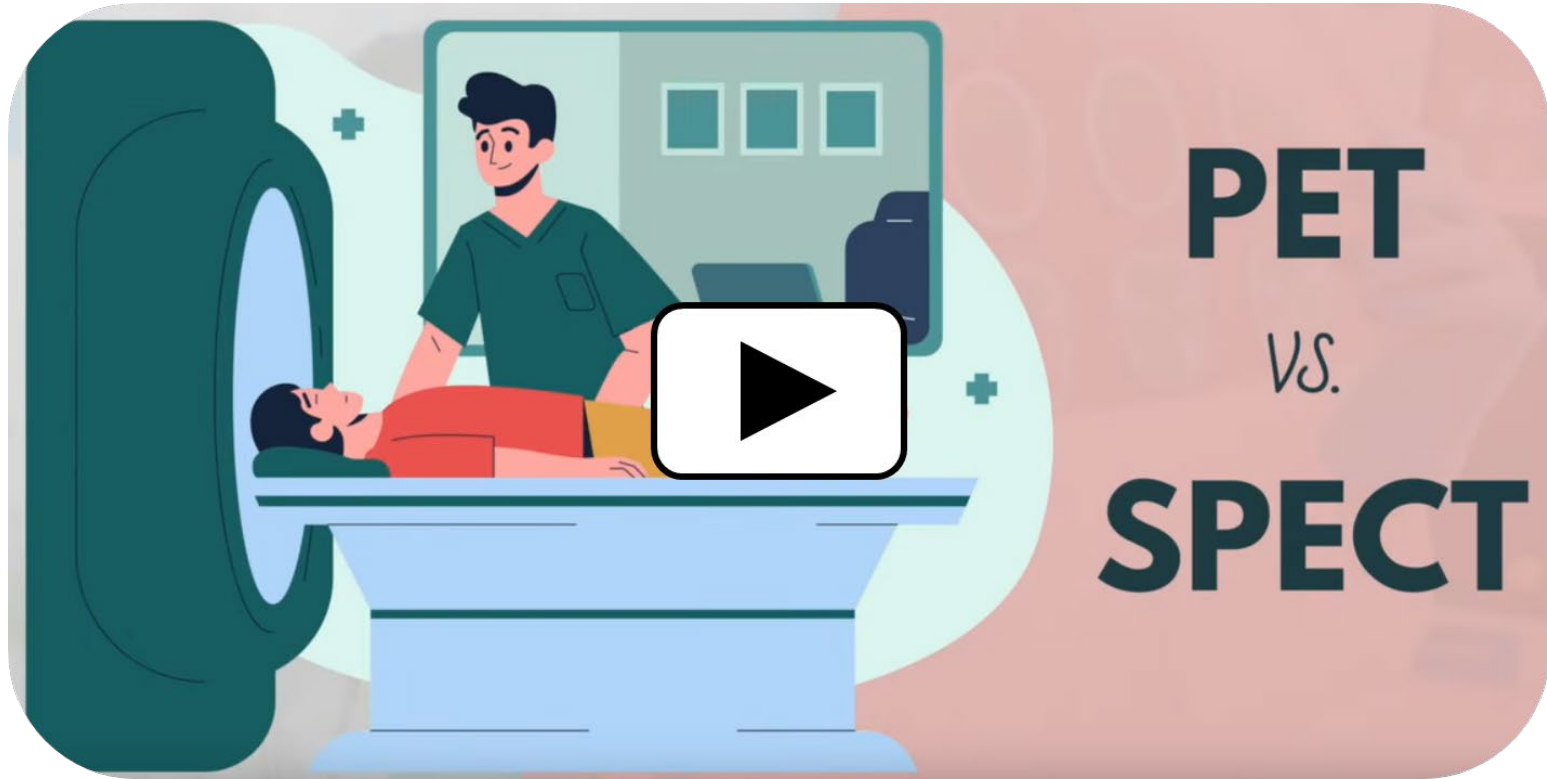
PET



PET

CT

PET/CT

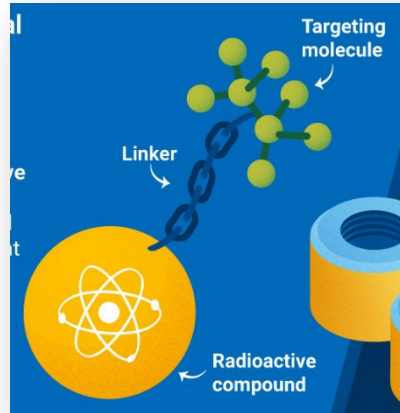


PET VS. SPECT

Radiotracers

Main principles of nuclear imaging: injection of a tiny amount of radioactive material («radiotracer»)

Specific tracers for specific tasks



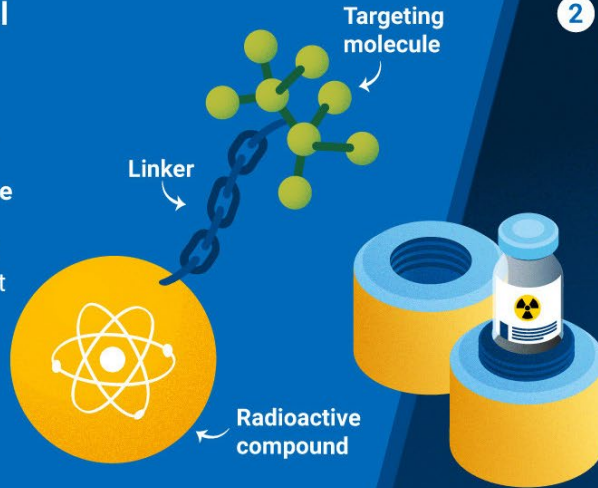
Radiotracer	Type of emission	Functional information (uptake mechanism)
^{99m}Tc -MIBI	Single-photon	Mitochondrial uptake
^{99m}Tc -maraciclatide	Single-photon	Angiogenesis
^{99m}Tc -annexin V	Single-photon	Apoptosis
^{99m}Tc -bombesin	Single-photon	Binding to BN receptor
^{123}I -labeled Z-MIVE	Single-photon	Binding to ER
^{18}F -FDG	Positron	Glucose metabolism
^{18}F -FMISO	Positron	Hypoxia
^{18}F -FLT	Positron	Proliferation
^{18}F -Galasco-RGS	Positron	Angiogenesis
^{18}F -annexin	Positron	Apoptosis
^{18}F -FES	Positron	Binding to ER
^{18}F -FFNP	Positron	Binding to PR
^{89}Zr -trastuzumab	Positron	Binding to HER2

^{99m}Tc -MIBI, technetium 99m-methoxyisobutylisonitrile; BN, growth factor bombesin; ^{123}I -labeled Z-MIVE, iodine 123 labeled cis-11 β -methoxy-17 α -iodovinyl estradiol; ER, estrogen receptor; ^{18}F -FDG, ^{18}F -fluorodeoxyglucose; ^{18}F -FMISO, ^{18}F -fluoromisonidazole; ^{18}F -FLT, ^{18}F -fluorothymidine; ^{18}F -Galacto-RGS, ^{18}F -galacto-recognizing arginine-glycine-aspartic acid; ^{18}F -FES, ^{18}F -fluoroestradiol; ^{18}F -FFNP, ^{18}F -fluoro furanyl norprogesterone; HER2, human epidermal growth factor receptor 2.

Radiotracers

1 Radiopharmaceutical Creation

Radioactive drugs, or radiopharmaceuticals, can be made by combining a **radioactive compound** that emits gamma rays, a tailored **targeting molecule** that varies to address specific organs or tissues, and a **linker** which ensures stable attachment.



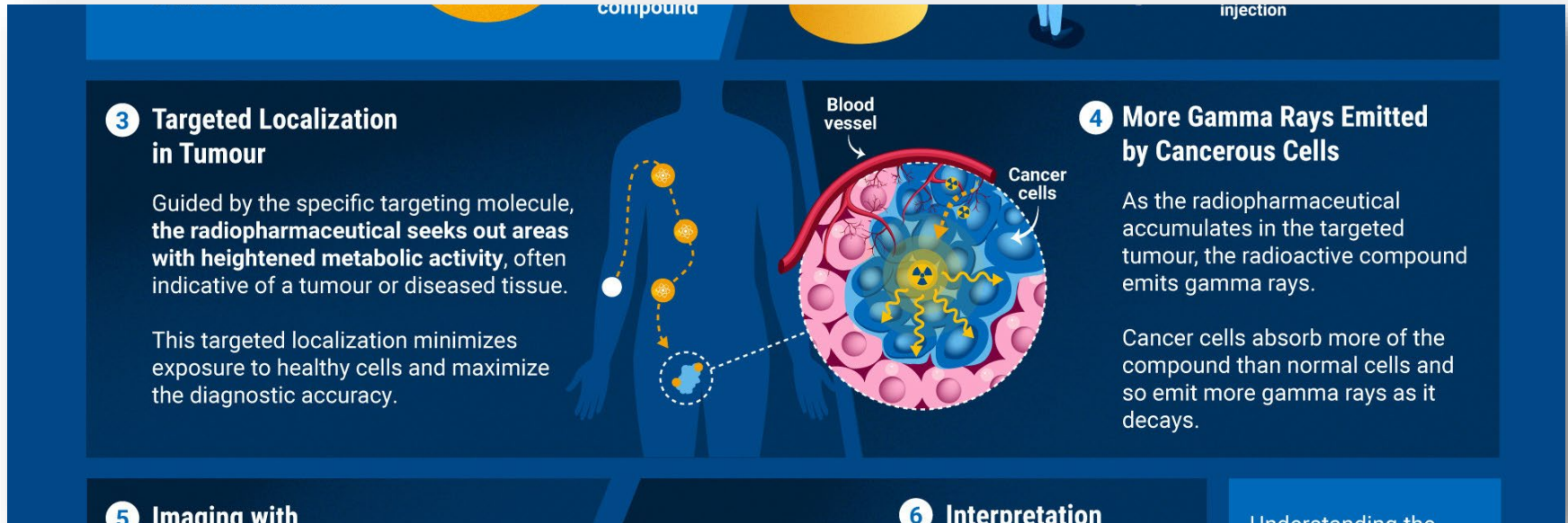
2 Administration into the Body

Patients receive the radiopharmaceutical through **injection or ingestion**. The method is selected based on the targeted organ or system.



source: iaea.org

Radiotracers

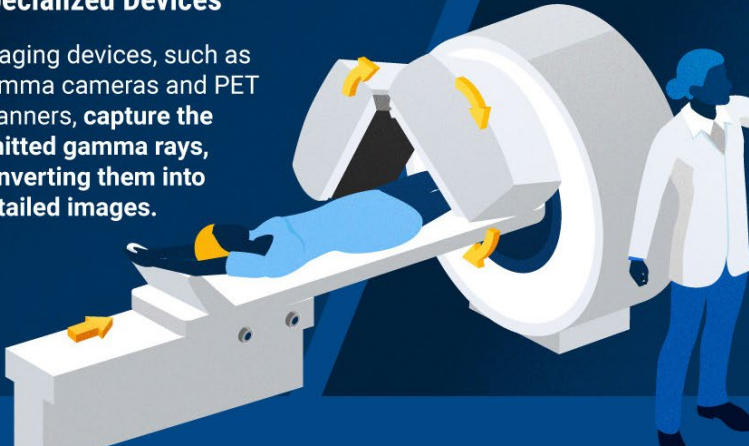


source: iaea.org

Radiotracers

5 Imaging with Specialized Devices

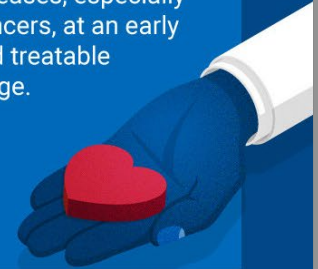
Imaging devices, such as gamma cameras and PET scanners, **capture the emitted gamma rays, converting them into detailed images.**



6 Interpretation and Diagnosis

Radiologists and nuclear medicine specialists interpret the obtained images to diagnose, locate abnormal cells and extent of diseases, facilitating accurate diagnosis.

Understanding the journey of radiopharmaceuticals within the body is pivotal for detection of diseases, especially cancers, at an early and treatable stage.



source: iaea.org

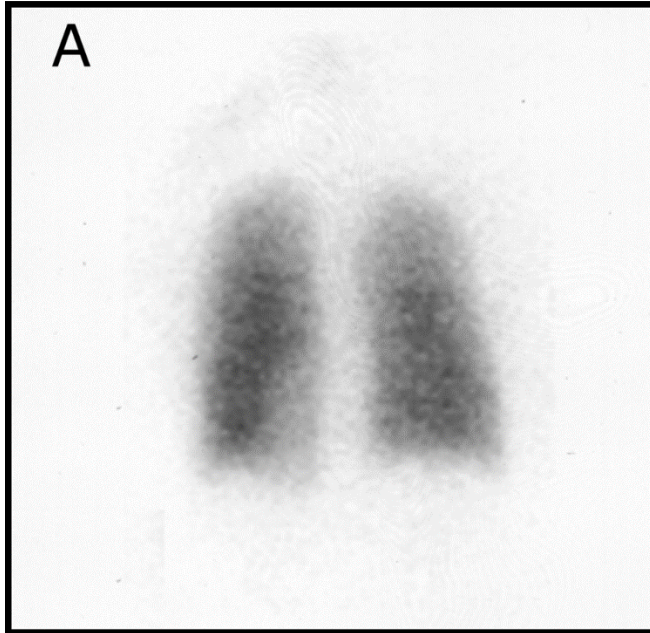
Content

- History
- Principles
- **Applications**
- Data Aspects

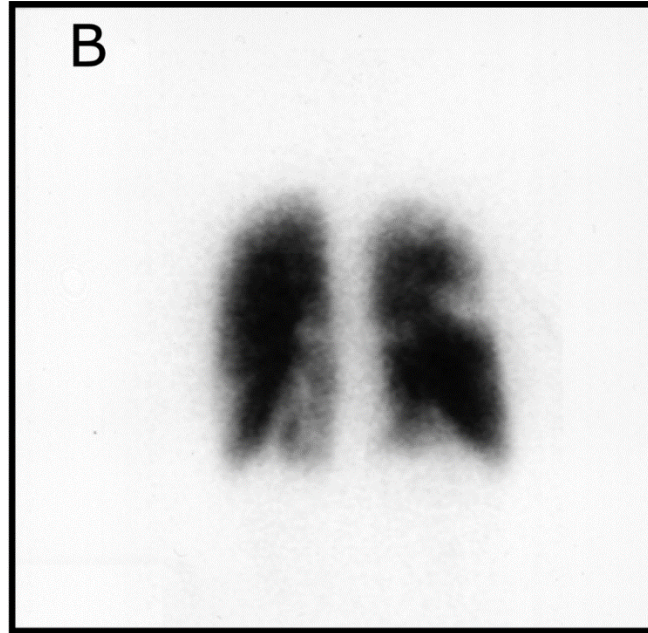


Clinical applications: Lung scintigraphy

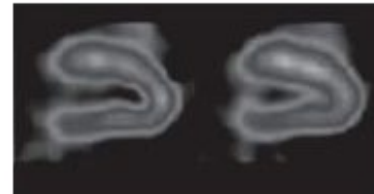
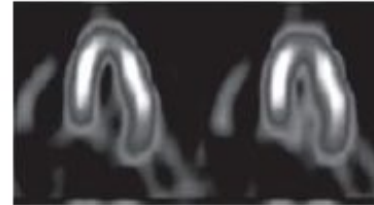
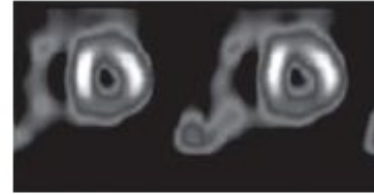
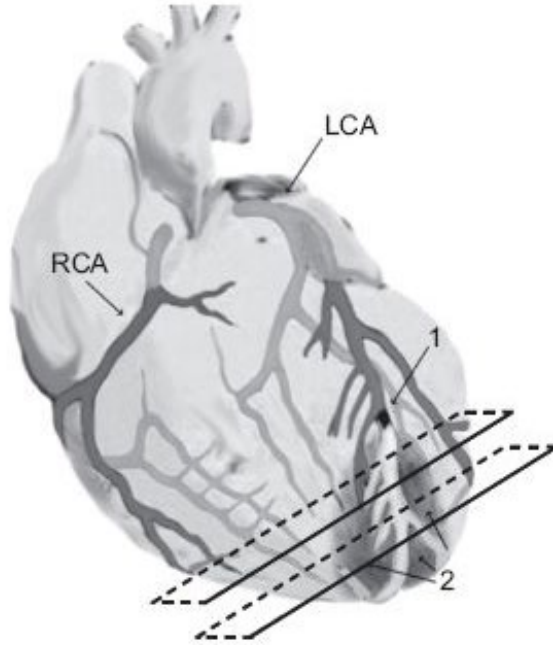
Ventilation scan:
inhalation of Xenon gas



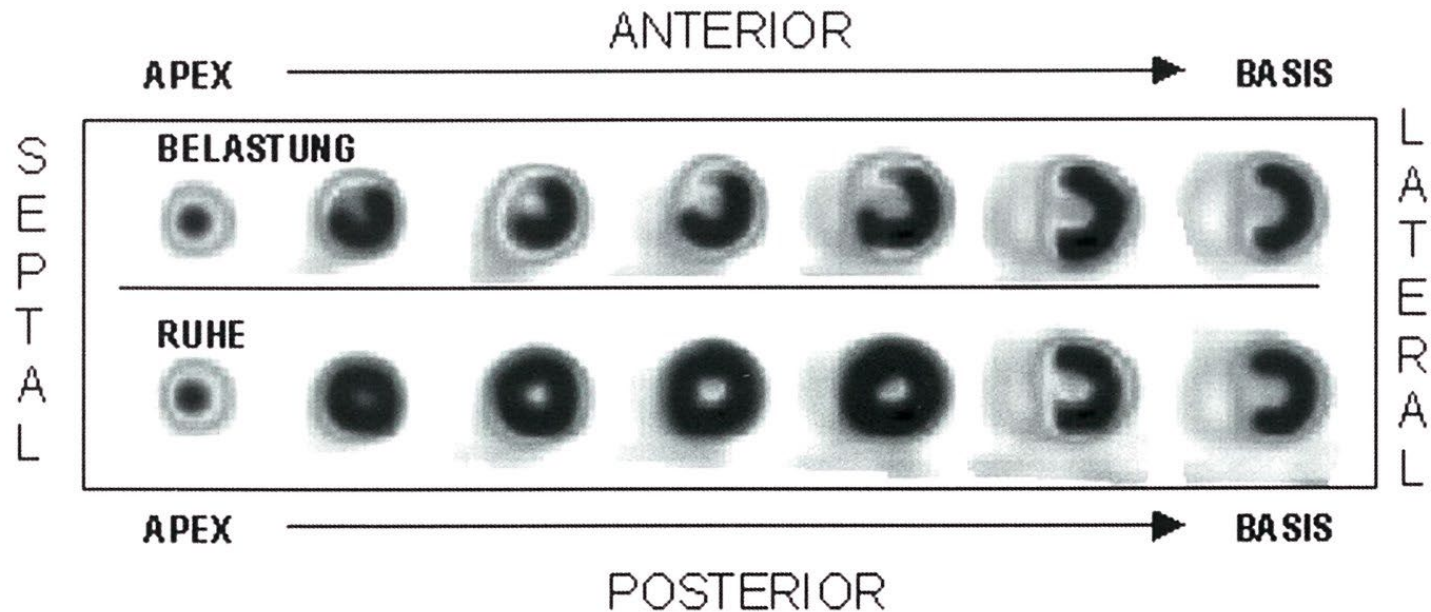
Perfusion scan:
intravenous injection of radiotracer



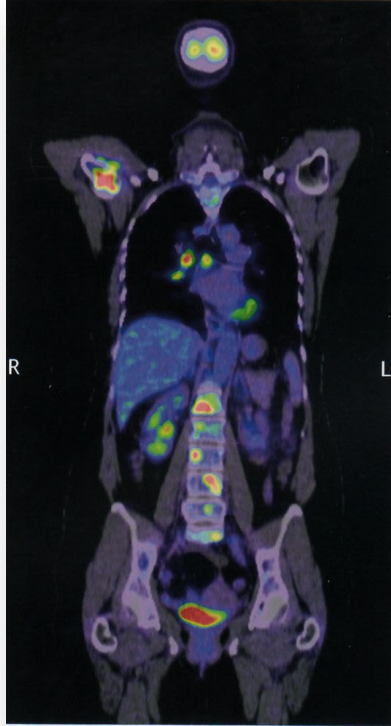
Clinical applications: Myocardial perfusion SPECT



Clinical applications: Myocardial perfusion SPECT



Clinical applications: Cancer PET



Content

- History
- Principles
- Applications
- **Data Aspects**



Image Properties

- **Usage:** infrequently used for several pathologies, because of special equipment and handling of radioactive substances
- **Projections:** planar images (scintigraphy) and tomographic (slice) images (SPECT & PET), latter nowadays fused with CT images
- **Artifacts:** radiotracers are highly specific, no background signal & high contrast
 - **Planar scintigraphy:** apparently different organ sizes (like X-ray images)
 - **PET & SPECT:** some organs always visible (e.g., bladder)
- **Noise:** relatively strong noise; visible: «grainy» images, because of
 - Radiotracer: tiny amount
 - Finite imaging time
- **Spatial resolution:** visible blurring of target on image, because of Collimator & gamma-camera (intrinsic)
 - Detector & post-processing (to enhance signal-to-noise ratio)

